

Transforming the Canonical Lagrangian of a Relativistic Particle for a Smooth Massless Limit: The Einbein Formulation in $3 + 1$ Dimensions

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Technical Tutorial Notes for Mathematical Physics Students (General Relativity, Lagrangian Mechanics, and the Massless Limit)

Abstract

This tutorial extends, step by step and without omitting intermediate algebraic manipulations, the einbein reformulation of the relativistic point-particle action from $1 + 1$ to $3 + 1$ spacetime dimensions. The variational core of the construction is dimension-independent and is re-derived in full: all Euler–Lagrange equations are computed explicitly, the algebraic solution for the einbein is substituted back into the action, and the recovery of the original square-root form is verified by direct calculation. The genuinely new content of the $3 + 1$ -dimensional setting is then isolated and developed: the null directions form the celestial sphere S^2 , the mass shell is the hyperbolic space H^3 , the massive little group is the rotation group $SO(3)$ with its $2s + 1$ -dimensional spin representations, the massless little group is the Euclidean group $ISO(2)$, and physical massless particles are characterized by a single helicity whose allowed values pair up as $\pm|\lambda|$ — so that the state counting jumps discontinuously at $m = 0$ even though the classical limit is perfectly smooth. The presentation is self-contained for a reader already familiar with the principles of general relativity, the Lagrangian formalism, and the Euler–Lagrange equations; familiarity with the companion $1 + 1$ -dimensional notes [0] is helpful but not assumed. References to standard textbooks and original papers are provided with author names, titles, publication years and, where available, stable URLs. Every derivable claim is backed by a machine check in the companion test suite (`tests/test_einbein_3plus1d_tutorial.py`), with test names mapped to equation numbers.

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1 Introduction and Motivation

The companion tutorial [0] carried out the einbein reformulation of the relativistic point particle in $1 + 1$ spacetime dimensions, deriving every algebraic step explicitly, and its $2 + 1$ -dimensional sequel re-derived the construction while isolating the first dimensional novelties (a circle of null directions, anyonic spin). The present notes perform the same construction in the physical case of $3 + 1$ dimensions. Here the massless limit acquires its full physical content:

- the null directions form the **celestial sphere** S^2 — the sky of an observer, in the literal sense;

- the massive little group is $SO(3)$, whose unitary representations are the familiar spin multiplets of dimension $2s + 1$, $s \in \{0, \frac{1}{2}, 1, \dots\}$;
- the massless little group is the two-dimensional Euclidean group $ISO(2)$; its translation subgroup must act trivially on physical states, and what remains is a single **helicity** λ , with $\lambda \in \{0, \pm\frac{1}{2}, \pm 1, \dots\}$ and the two values $\pm|\lambda|$ paired for any massless field;
- consequently the massless limit, while smooth at the level of the action and of the geodesics, **cannot** be smooth at the level of state counting: a massive spin- s particle carries $2s + 1$ polarization states, a massless one carries two. For interacting theories this kinematic fact becomes a genuine discontinuity of physical observables (the van Dam–Veltman–Zakharov phenomenon for massive gravity, Section 10.4).

The einbein device itself is indifferent to the dimension: the same auxiliary world-line field $e(\tau) > 0$ converts the square-root action into a quadratic one, the same algebraic constraint appears, and the same reparametrization gauge $e = \text{const}$ restores the affine geodesic equation. Sections 2–9 re-derive the construction with all steps displayed, so that the present document remains self-contained in the style of [0]. Sections 10–12 develop the dimensional novelties.

A roadmap for the reader who has studied [0]: Sections 2–9 are the $1 + 1$ derivation with four indices instead of two; the only places where the dimension shows are the counting statements (forty independent Christoffel components, listed in full in Appendix A) and Sections 9.3, 10, and 12, which contain the new material.

2 Notation and Geometric Setup in $3 + 1$ Dimensions

We work on a four-dimensional Lorentzian manifold M equipped with a metric tensor $g_{\mu\nu}$ of signature $(-, +, +, +)$. Local coordinates are denoted $x^\mu = (x^0, x^1, x^2, x^3)$. A world-line is a smooth map

$$x^\mu : I \subset \mathbb{R} \longrightarrow M, \quad \tau \longmapsto x^\mu(\tau), \quad (1)$$

where τ is an arbitrary parameter (not necessarily proper time). Derivatives with respect to τ are written with a dot: $\dot{x}^\mu(\tau) := \frac{dx^\mu}{d\tau}$.

The induced interval on the world-line is

$$ds^2 = g_{\mu\nu}(x(\tau)) dx^\mu dx^\nu = g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu d\tau^2. \quad (2)$$

We adopt the convention that a future-directed timelike vector v^μ satisfies

$$g_{\mu\nu} v^\mu v^\nu < 0. \quad (3)$$

Consequently the quantity

$$\sigma := g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu \quad (4)$$

is negative for a timelike world-line. Because the signature is $(-, +, +, +)$, the metric has one negative and three positive eigenvalues; the determinant, being the product of the eigenvalues, is negative:

$$\det(g_{\mu\nu}) < 0. \quad (5)$$

(The Leibniz expansion of the 4×4 determinant has $4! = 24$ terms; unlike the $1 + 1$ and $2 + 1$ cases we do not print it — the sign follows from the eigenvalue argument alone, and the full expansion is never needed below. The eigenvalue argument itself is checked numerically in the companion suite.)

Throughout we keep the speed of light c_0 explicit. Rest mass is denoted $m \geq 0$.

3 The Canonical Square-Root Action

The action that extremizes proper time is

$$S_{\text{sqr}}[x] = -mc_0 \int_{\tau_i}^{\tau_f} \sqrt{-g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu} d\tau = -mc_0 \int \sqrt{-\sigma} d\tau. \quad (6)$$

The associated Lagrangian density (with respect to the parameter τ) is therefore

$$L_{\text{sqr}} = -mc_0 \sqrt{-g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} = -mc_0 \sqrt{-\sigma}. \quad (7)$$

Because L_{sqr} is homogeneous of degree one in the velocities, the action is invariant under arbitrary reparametrizations $\tau \rightarrow f(\tau)$ with $\dot{f} > 0$.¹ The Euler–Lagrange equations obtained from L_{sqr} are equivalent to the geodesic equation

$$\frac{D\dot{x}^\mu}{d\tau} = 0 \quad (8)$$

when the parameter is chosen to be an affine parameter² (or, more generally, proportional to proper time). None of these statements involves the dimension of M ; the proofs given in [0], Section 3, apply verbatim.

4 Difficulties with the Massless Limit

If one attempts to set $m = 0$ directly in (7), the action vanishes for every trajectory that satisfies the null condition $\sigma = 0$. The variational principle then ceases to select a preferred class of curves: every null curve yields the same (zero) value of the action. Moreover, the square-root Lagrangian becomes non-differentiable on the light cone. Consequently a different formulation is required if one wishes a single action principle that covers both the massive and the massless cases continuously.

In $3 + 1$ dimensions the kinematic structure behind this difficulty is richer than in lower dimensions. For a massive particle the tangent vector lives in the interior of the future light cone; as $m \rightarrow 0$ it is pushed onto the cone’s boundary, whose space of directions is a two-sphere (Section 9.3). The fate of the particle’s *internal* rotational state in that limit is the content of Wigner’s little-group analysis, developed in Section 10: it is there that the smooth classical limit and the non-smooth quantum state counting are reconciled.

5 The Einbein Reformulation

Introduce a real auxiliary field $e(\tau) > 0$, called the einbein. Geometrically, e may be viewed as defining an intrinsic metric on the one-dimensional world-line via

$$ds_{\text{worldline}}^2 = e(\tau)^2 d\tau^2. \quad (9)$$

Consider the new action functional that depends on both the embedding $x^\mu(\tau)$ and the einbein $e(\tau)$:

$$S[x, e] = \int_{\tau_i}^{\tau_f} \left[\frac{1}{2e} g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu - \frac{e}{2} (mc_0)^2 \right] d\tau. \quad (10)$$

¹“Reparametrization invariance” means that the value of the action does not change if the world-line is relabeled by any strictly increasing function f : the curve as a geometric object is unchanged, only its parametrization. Homogeneity of degree one, $L(x, \lambda \dot{x}) = \lambda L(x, \dot{x})$ for $\lambda > 0$, is precisely the property that makes the integral parameter-independent by the change-of-variables formula.

²A parameter τ along a geodesic is called affine if the tangent vector is parallel-transported without rescaling: $D\dot{x}^\mu/d\tau = 0$. Any other parameter produces an extra term proportional to $\dot{x}^\mu$; such a parameter is called non-affine. Affine parameters along a given geodesic are unique up to affine changes $\tau \rightarrow a\tau + b$; see S. M. Carroll, *Spacetime and Geometry* (Cambridge University Press, 2019), §3.3.

The corresponding Lagrangian is

$$L(x, \dot{x}, e) = \frac{1}{2e} \sigma - \frac{e}{2} (mc_0)^2, \quad (11)$$

where $\sigma = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$ as before. Note that L is quadratic in the velocities and contains no derivative of e ; the einbein is therefore an auxiliary (non-dynamical) field.³ The action (10) is invariant under the reparametrizations

$$\tau \mapsto f(\tau), \quad e(\tau) \mapsto \frac{e(\tau)}{\dot{f}(\tau)} \quad (\dot{f} > 0), \quad (12)$$

because under the combined transformation the first term of the integrand picks up $1/\dot{f}^2$ from the two velocities and $\dot{f}$ from the measure, for a net $1/\dot{f}$ — which is exactly cancelled by the einbein's law $1/e \rightarrow \dot{f}/e$; the second term picks up $1/\dot{f}$ from $e \rightarrow e/\dot{f}$ and $\dot{f}$ from the measure, and is invariant as it stands. The invariant combination is $e(\tau) d\tau$, the world-line volume element $\sqrt{g_{\text{worldline}}} d\tau$ of (9). The einbein is thus not an optional trick but the required compensating field: it is the degree-one object on the world-line whose transformation law absorbs the Jacobian $\dot{f}$ so that a quadratic action can remain reparametrization-invariant.

6 Euler–Lagrange Equation for the Einbein

The general Euler–Lagrange equation associated with any variable $\varphi(\tau)$ is

$$\frac{d}{d\tau} \frac{\partial L}{\partial \dot{\varphi}} - \frac{\partial L}{\partial \varphi} = 0. \quad (13)$$

We now specialise this equation to the einbein by a sequence of purely logical steps, exactly as in [0], Section 6.

Step 1. Start from the general Euler–Lagrange equation (13).

Step 2. Choose the variable $\varphi = e(\tau)$. The equation becomes

$$\frac{d}{d\tau} \frac{\partial L}{\partial \dot{e}} - \frac{\partial L}{\partial e} = 0. \quad (14)$$

Step 3. Inspection of the explicit Lagrangian

$$L = \frac{1}{2e} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu - \frac{e}{2} (mc_0)^2 \quad (15)$$

shows that the derivative $\dot{e}$ never appears. Consequently the partial derivative with respect to that velocity vanishes identically:

$$\frac{\partial L}{\partial \dot{e}} \equiv 0. \quad (16)$$

Step 4. The ordinary derivative of the zero function is zero:

$$\frac{d}{d\tau} \frac{\partial L}{\partial \dot{e}} = \frac{d}{d\tau} (0) = 0. \quad (17)$$

³An auxiliary (or non-dynamical) field is one whose equation of motion contains no derivatives of the field itself and can therefore be solved algebraically in terms of the other fields. Substituting the algebraic solution back into the action returns an equivalent description of the same classical theory.

Step 5. Substituting the result of Step 4 into the specialised equation of Step 2 leaves only

$$-\frac{\partial L}{\partial e} = 0. \quad (18)$$

Step 6. Multiplying both sides by -1 yields the equivalent statement

$$\frac{\partial L}{\partial e} = 0. \quad (19)$$

Thus equation (19) is nothing but the general Euler–Lagrange equation after the observation that $\partial L/\partial \dot{e}$ is identically zero has been used. No further dynamical assumption is required; the reduction is purely formal.

We now compute the partial derivative $\partial L/\partial e$ term by term. The Lagrangian is the sum of two pieces:

$$L = \underbrace{\frac{1}{2e} \sigma}_{L_1} + \underbrace{\left(-\frac{e}{2}(mc_0)^2\right)}_{L_2}.$$

Differentiating each piece separately with respect to e (treating the velocity combination σ as independent of e) yields

$$\frac{\partial L_1}{\partial e} = \frac{\partial}{\partial e} \left(\frac{1}{2e} \sigma \right) = \sigma \cdot \frac{\partial}{\partial e} \left(\frac{1}{2e} \right) = \sigma \cdot \left(-\frac{1}{2e^2} \right) = -\frac{1}{2e^2} \sigma, \quad (20)$$

$$\frac{\partial L_2}{\partial e} = \frac{\partial}{\partial e} \left(-\frac{e}{2}(mc_0)^2 \right) = -\frac{1}{2}(mc_0)^2. \quad (21)$$

Because differentiation is a linear operation,

$$\frac{\partial L}{\partial e} = \frac{\partial L_1}{\partial e} + \frac{\partial L_2}{\partial e} = -\frac{1}{2e^2} \sigma - \frac{1}{2}(mc_0)^2. \quad (22)$$

The Euler–Lagrange condition $\partial L/\partial e = 0$ (already established in equation (19)) therefore requires

$$-\frac{1}{2e^2} \sigma - \frac{1}{2}(mc_0)^2 = 0. \quad (23)$$

To solve the algebraic equation we multiply both sides by the nowhere-vanishing factor $-2e^2$:

$$(-2e^2) \left(-\frac{1}{2e^2} \sigma \right) + (-2e^2) \left(-\frac{1}{2}(mc_0)^2 \right) = 0 \implies \sigma + e^2(mc_0)^2 = 0. \quad (24)$$

Equivalently,

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -e^2(mc_0)^2. \quad (25)$$

Solving for the positive root $e > 0$ gives

$$e(\tau) = \frac{\sqrt{-\sigma}}{mc_0} = \frac{\sqrt{-g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu}}{mc_0}. \quad (26)$$

(The negative root would correspond to a time-reversed parametrization and is discarded by the orientation convention $e > 0$.)

Observe that the dimension of the spacetime has entered this section only through the index range of μ in σ . Every step is valid for any dimension, in particular for $3 + 1$.

7 Substitution and Recovery of the Square-Root Action

We now substitute the algebraic solution (26) back into the Lagrangian (11) and verify that the original square-root Lagrangian is recovered exactly.

From the constraint (25) we have

$$\sigma = -e^2(mc_0)^2. \quad (27)$$

Insert this expression into the first term of L :

$$\frac{1}{2e}\sigma = \frac{-e^2(mc_0)^2}{2e} = -\frac{e}{2}(mc_0)^2. \quad (28)$$

Therefore the whole Lagrangian becomes

$$L = -\frac{e}{2}(mc_0)^2 - \frac{e}{2}(mc_0)^2 = -e(mc_0)^2. \quad (29)$$

Now replace e by its explicit solution (26):

$$L = -\left(\frac{\sqrt{-\sigma}}{mc_0}\right)(mc_0)^2 = -mc_0\sqrt{-\sigma} = -mc_0\sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}. \quad (30)$$

This is precisely L_{sqrt} of equation (7). Consequently the two action principles are classically equivalent for $m > 0$: every critical point of $S[x, e]$ that satisfies the einbein equation of motion projects to a critical point of $S_{\text{sqrt}}[x]$, and vice versa. The equivalence holds in any spacetime dimension; the $3 + 1$ -dimensional case needed no modification.

8 Euler–Lagrange Equations for the Coordinates (Massive Case)

We next derive the equations of motion for the embedding coordinates $x^\lambda(\tau)$ in the massive case $m > 0$. The Lagrangian

$$L(x, \dot{x}, e) = \frac{1}{2e}g_{\mu\nu}(x)\dot{x}^\mu\dot{x}^\nu - \frac{e}{2}(mc_0)^2 \quad (31)$$

depends on the coordinates both through the metric coefficients $g_{\mu\nu}(x)$ and through the velocities $\dot{x}^\mu$. The Euler–Lagrange equation for each component x^λ ($\lambda = 0, 1, 2, 3$) is

$$\frac{d}{d\tau}\frac{\partial L}{\partial \dot{x}^\lambda} = \frac{\partial L}{\partial x^\lambda}. \quad (32)$$

Throughout this section the einbein $e(\tau)$ is treated as an independent field (its own equation of motion will be imposed afterwards).

8.1 Partial derivative with respect to the velocity

Differentiate L with respect to $\dot{x}^\lambda$. Only the first term contributes:

$$\frac{\partial L}{\partial \dot{x}^\lambda} = \frac{\partial}{\partial \dot{x}^\lambda} \left(\frac{1}{2e} g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu \right) = \frac{1}{2e} g_{\mu\nu}(x) (\delta^\mu_\lambda \dot{x}^\nu + \dot{x}^\mu \delta^\nu_\lambda) = \frac{1}{2e} (g_{\lambda\nu} \dot{x}^\nu + g_{\mu\lambda} \dot{x}^\mu) = \frac{1}{e} g_{\lambda\nu}(x) \dot{x}^\nu, \quad (33)$$

where the last equality follows from the symmetry $g_{\mu\lambda} = g_{\lambda\mu}$. Consequently the left-hand side of the Euler–Lagrange equation is the total τ -derivative

$$\frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}^\lambda} = \frac{d}{d\tau} \left(\frac{1}{e} g_{\lambda\nu}(x) \dot{x}^\nu \right). \quad (34)$$

8.2 Partial derivative with respect to the coordinate

At fixed velocities the only dependence of L on the coordinate x^λ comes from the metric tensor:

$$\frac{\partial L}{\partial x^\lambda} = \frac{1}{2e} (\partial_\lambda g_{\mu\nu}(x)) \dot{x}^\mu \dot{x}^\nu. \quad (35)$$

(The second term of L does not depend on x .)

8.3 The Euler–Lagrange equation before imposing the einbein constraint

Equating (34) and (35) yields the explicit Euler–Lagrange equation

$$\frac{d}{d\tau} \left(\frac{1}{e} g_{\lambda\nu} \dot{x}^\nu \right) = \frac{1}{2e} (\partial_\lambda g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu. \quad (36)$$

We now convert (36) into an equivalent form by a sequence of elementary calculus steps.

Step 1. The left-hand side of (36) is the ordinary derivative of a product of two τ -dependent factors, $\frac{1}{e(\tau)}$ and $g_{\lambda\nu}(x(\tau)) \dot{x}^\nu(\tau)$. Apply the ordinary product rule:

$$\frac{d}{d\tau} \left(\frac{1}{e} g_{\lambda\nu} \dot{x}^\nu \right) = \left(\frac{d}{d\tau} \frac{1}{e} \right) (g_{\lambda\nu} \dot{x}^\nu) + \frac{1}{e} \frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu). \quad (37)$$

Step 2. Differentiate the reciprocal of the einbein by the chain rule:

$$\frac{d}{d\tau} \frac{1}{e} = \frac{d}{d\tau} (e^{-1}) = -e^{-2} \dot{e} = -\frac{\dot{e}}{e^2}. \quad (38)$$

Step 3. Substitute the result of Step 2 back into the product rule of Step 1:

$$\frac{d}{d\tau} \left(\frac{1}{e} g_{\lambda\nu} \dot{x}^\nu \right) = -\frac{\dot{e}}{e^2} (g_{\lambda\nu} \dot{x}^\nu) + \frac{1}{e} \frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu). \quad (39)$$

Step 4. Equation (36) therefore reads

$$-\frac{\dot{e}}{e^2} g_{\lambda\nu} \dot{x}^\nu + \frac{1}{e} \frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu) = \frac{1}{2e} (\partial_\lambda g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu. \quad (40)$$

Step 5. Multiply every term of the equation by the positive factor e (which never vanishes by definition of the einbein). The first term becomes $-\frac{\dot{e}}{e} g_{\lambda\nu} \dot{x}^\nu$, the second term becomes $\frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu)$, and the right-hand side becomes $\frac{1}{2} (\partial_\lambda g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu$.

Step 6. Rearrange the resulting terms to obtain

$$\frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu) - \frac{\dot{e}}{e} g_{\lambda\nu} \dot{x}^\nu = \frac{1}{2} (\partial_\lambda g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu. \quad (41)$$

All six steps are ordinary calculus (product rule and chain rule) together with multiplication by a nowhere-vanishing positive function; no geometric identity is required for this particular transition, and no step refers to the spacetime dimension.

8.4 Expansion of the total derivative and introduction of the Christoffel symbols

Expand the total derivative that appears on the left-hand side of (41). Write the product as $g_{\lambda\nu}(x(\tau)) \dot{x}^\nu(\tau)$ and apply the ordinary product rule of one-variable calculus:

$$\frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu) = \left(\frac{d}{d\tau} g_{\lambda\nu} \right) \dot{x}^\nu + g_{\lambda\nu} \ddot{x}^\nu. \quad (42)$$

The first term requires care because the metric components are functions of the spacetime coordinates, which themselves depend on τ . In $3 + 1$ dimensions this dependence may be written explicitly:

$$g_{\lambda\nu} = g_{\lambda\nu}(x^0(\tau), x^1(\tau), x^2(\tau), x^3(\tau)).$$

The ordinary chain rule of multivariable calculus then states

$$\frac{d}{d\tau} g_{\lambda\nu} = \frac{\partial g_{\lambda\nu}}{\partial x^0} \dot{x}^0 + \frac{\partial g_{\lambda\nu}}{\partial x^1} \dot{x}^1 + \frac{\partial g_{\lambda\nu}}{\partial x^2} \dot{x}^2 + \frac{\partial g_{\lambda\nu}}{\partial x^3} \dot{x}^3, \quad (43)$$

abbreviated in index notation as

$$\frac{d}{d\tau} g_{\lambda\nu} = (\partial_\rho g_{\lambda\nu}) \dot{x}^\rho, \quad (44)$$

with ρ summed over $0, 1, 2, 3$. Multiplying by the remaining velocity $\dot{x}^\nu$ produces the quadratic term

$$\left(\frac{d}{d\tau} g_{\lambda\nu} \right) \dot{x}^\nu = (\partial_\rho g_{\lambda\nu}) \dot{x}^\rho \dot{x}^\nu. \quad (45)$$

Collecting both contributions yields the expansion

$$\frac{d}{d\tau} (g_{\lambda\nu} \dot{x}^\nu) = (\partial_\rho g_{\lambda\nu}) \dot{x}^\rho \dot{x}^\nu + g_{\lambda\nu} \ddot{x}^\nu. \quad (46)$$

Substitution of this expansion into (41) produces

$$(\partial_\rho g_{\lambda\nu}) \dot{x}^\rho \dot{x}^\nu + g_{\lambda\nu} \ddot{x}^\nu - \frac{\dot{e}}{e} g_{\lambda\nu} \dot{x}^\nu = \frac{1}{2} (\partial_\lambda g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu. \quad (47)$$

Raising the free index λ proceeds exactly as in the $1 + 1$ and $2 + 1$ cases: multiply by the inverse metric $g^{\sigma\lambda}$, distribute term by term, use $g^{\sigma\lambda} g_{\lambda\nu} = \delta^\sigma_\nu$ on the acceleration and einbein terms, and symmetrize the

metric-derivative combination using the symmetry of the velocity bilinear $\dot{x}^\rho \dot{x}^\nu$ (renaming $\rho \leftrightarrow \nu$ shows the two placements of the first derivative term are equal). The result is

$$\ddot{x}^\sigma + \frac{1}{2} g^{\sigma\lambda} (\partial_\rho g_{\lambda\nu} + \partial_\nu g_{\lambda\rho} - \partial_\lambda g_{\rho\nu}) \dot{x}^\rho \dot{x}^\nu - \frac{\dot{e}}{e} \dot{x}^\sigma = 0. \quad (48)$$

The combination of metric derivatives in (48) is the Christoffel symbol of the Levi-Civita connection,

$$\Gamma^\sigma_{\rho\nu} = \frac{1}{2} g^{\sigma\lambda} (\partial_\rho g_{\lambda\nu} + \partial_\nu g_{\lambda\rho} - \partial_\lambda g_{\rho\nu}). \quad (49)$$

Because the lower indices are symmetric, the number of independent components in n spacetime dimensions is $n^2(n+1)/2$: six in $1+1$ dimensions, eighteen in $2+1$, and **forty in $3+1$** . All forty are listed explicitly in Appendix A; the listing there is generated programmatically from (49) and is checked component-by-component against (49) in the companion test suite, so the appendix can be trusted as a lookup table in explicit calculations.

With these symbols the Euler–Lagrange equation takes the compact form

$$\ddot{x}^\sigma + \Gamma^\sigma_{\rho\nu} \dot{x}^\rho \dot{x}^\nu - \frac{\dot{e}}{e} \dot{x}^\sigma = 0, \quad (50)$$

or, equivalently,

$$\frac{D}{d\tau} \left(\frac{\dot{x}^\sigma}{e} \right) = 0, \quad (51)$$

where $\frac{D}{d\tau}$ denotes the covariant derivative along the world-line,

$$\frac{DV^\sigma}{d\tau} := \frac{dV^\sigma}{d\tau} + \Gamma^\sigma_{\rho\nu} \dot{x}^\rho V^\nu \quad (52)$$

for any vector field V^σ defined along the curve.

Equation (50) is the geodesic equation with respect to a non-affine parameter; the term proportional to $\dot{e}/e$ measures the failure of τ to be affine. The resolution is dimension-independent and proceeds in the five standard steps: (1) the residual reparametrization freedom (12) is still unfixed; (2) the einbein equation of motion imposes the algebraic constraint (25); (3) the freedom is spent on the gauge condition⁴

$$e(\tau) = \text{const} \implies \dot{e} = 0; \quad (53)$$

(4) in that gauge the non-affine term vanishes identically and (50) collapses to the affine geodesic equation

$$\frac{D\dot{x}^\sigma}{d\tau} = \ddot{x}^\sigma + \Gamma^\sigma_{\rho\nu} \dot{x}^\rho \dot{x}^\nu = 0; \quad (54)$$

(5) the constraint (25) becomes the tangent-vector normalization

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -e^2 (mc_0)^2 = \text{const} < 0, \quad (55)$$

identifying τ with a multiple of the proper time.

⁴In the language of constrained systems a gauge condition is an extra equation that one is free to impose in order to remove the redundancy associated with a local symmetry. It is not an equation of motion; it is a choice of representative inside each equivalence class of reparametrizations.

Thus the Euler–Lagrange equations of the einbein action, after the einbein constraint is taken into account and a convenient gauge is chosen, are exactly the geodesic equations of the background metric — in $3 + 1$ dimensions exactly as in $1 + 1$ and $2 + 1$.

9 The Massless Case: Euler–Lagrange Equations and Null Geodesics

We now set the rest mass to zero, $m = 0$. The Lagrangian simplifies to

$$L = \frac{1}{2e} g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu. \quad (56)$$

9.1 Euler–Lagrange equation for the einbein (massless)

Because L still does not depend on $\dot{e}$, the equation $\partial L / \partial e = 0$ reduces to

$$-\frac{1}{2e^2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0, \quad (57)$$

which immediately implies the null constraint

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0. \quad (58)$$

9.2 Euler–Lagrange equations for the coordinates (massless)

The partial derivatives with respect to velocity and coordinate are identical in form to those of the massive case (the mass term is absent, but it never contributed to these derivatives). Consequently one obtains again (36), and the same algebraic steps produce

$$\ddot{x}^\sigma + \Gamma^\sigma_{\rho\nu} \dot{x}^\rho \dot{x}^\nu - \frac{\dot{e}}{e} \dot{x}^\sigma = 0, \quad (59)$$

with the Christoffel symbols of (49) (expanded in Appendix A). The residual reparametrization freedom is identical, the gauge $e = \text{const}$ is still available, and in that gauge (59) collapses to the affine null-geodesic equation

$$\frac{D\dot{x}^\sigma}{d\tau} = 0, \quad g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0. \quad (60)$$

The massless limit is therefore completely regular: the action with Lagrangian (56) is well-defined, the constraint (58) is enforced dynamically by the variation of the einbein, and the resulting trajectories are exactly the null geodesics.

9.3 What the null constraint does and does not fix in $3 + 1$ dimensions

Write the null constraint (58) in a locally inertial frame at a point, where $g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$:

$$-(\dot{x}^0)^2 + (\dot{x}^1)^2 + (\dot{x}^2)^2 + (\dot{x}^3)^2 = 0. \quad (61)$$

For a future-directed world-line $\dot{x}^0 > 0$, and all solutions are parametrized by a point of the unit two-sphere:

$$\dot{x}^1 = \dot{x}^0 \sin \vartheta \cos \varphi, \quad \dot{x}^2 = \dot{x}^0 \sin \vartheta \sin \varphi, \quad \dot{x}^3 = \dot{x}^0 \cos \vartheta, \quad (\vartheta, \varphi) \in S^2. \quad (62)$$

The null constraint is **one** equation for the **three** independent ratios of the four velocity components, leaving a continuous two-parameter family of null directions: the sphere S^2 . This is the celestial sphere of the

observer at the point in question — the set of directions on the sky from which light can arrive. In n spacetime dimensions the space of null directions is S^{n-2} : two points in $1 + 1$, a circle in $2 + 1$, a sphere in $3 + 1$.

The practical consequence for the initial-value problem: the direction (ϑ_0, φ_0) of the initial null vector is continuous two-dimensional initial data that the equations of motion cannot supply.

10 Where the Dimension Enters: The Celestial Sphere, Mass Shells, and Little Groups

This section develops the structural facts that give the $3 + 1$ -dimensional massless limit its physical content. Everything here is standard; the original references are Wigner’s 1939 classification and Weinberg’s textbook account, and the checks that admit a machine form are in the companion suite.

10.1 The mass shell as a homogeneous space

Work in flat $3 + 1$ -dimensional Minkowski space with the momentum $p_\mu = (p_0, p_1, p_2, p_3)$ and mass shell (derived in Section 11 below)

$$g^{\mu\nu} p_\mu p_\nu = -(mc_0)^2. \quad (63)$$

For $m > 0$ the future sheet of this two-sheeted hyperboloid is the hyperbolic three-space H^3 : the restriction of the Minkowski quadratic form to the shell’s tangent spaces is positive-definite (the hyperboloid model; J. G. Ratcliffe, *Foundations of Hyperbolic Manifolds*, Springer, 2nd ed. 2006, Chapter 3 — verified numerically in the companion suite). The Lorentz group $SO(3, 1)$ acts transitively⁵ on this sheet: any future-directed timelike p can be rotated and boosted to the rest-frame value $p^{(0)} = (mc_0, 0, 0, 0)$. The stabilizer of $p^{(0)}$ consists of the spatial rotations: the **little group**⁶ of a massive particle in $3 + 1$ dimensions is

$$G_{\text{massive}} = SO(3). \quad (64)$$

For $m = 0$ the shell degenerates to the future null cone with the origin removed. The Lorentz group still acts transitively (any future-null vector can be brought to the standard form $k^{(0)} = (\kappa, 0, 0, \kappa)$, $\kappa > 0$), but the stabilizer is now three-dimensional: it is generated by the rotation about the direction of motion and by two null rotations. With the conventions fixed in the companion suite (boosts K_i with $(K_i)^0_i = (K_i)^i_0 = 1$, rotations J_i with $(J_3)^1_2 = +1 = -(J_3)^2_1$ and cyclic images, matrices acting on $(p_0, p_1, p_2, p_3)^T$), the three generators are

$$T_1 := K_1 + J_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad T_2 := K_2 - J_1 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (65)$$

and one verifies by direct multiplication that

$$T_1 k^{(0)} = T_2 k^{(0)} = J_3 k^{(0)} = 0, \quad (66)$$

so all three fix the standard null vector, and their exponentials are Lorentz transformations (each ηM is antisymmetric). Their commutators, again by direct multiplication, are

⁵A group G acts transitively on a set X if any point of X can be carried to any other by some group element. The set X is then a single orbit, identifiable with the coset space G/G_x for the stabilizer G_x of any chosen point x .

⁶The little group (or stabilizer) of a particle’s momentum p is the subgroup of the Lorentz group that leaves p unchanged. Wigner’s classification (1939) identifies the internal states of an elementary particle with the irreducible unitary representations of the little group of its momentum.

$$[J_3, T_1] = -T_2, \quad [J_3, T_2] = +T_1, \quad [T_1, T_2] = 0, \quad (67)$$

which is precisely the Lie algebra of the Euclidean group of the plane: T_1, T_2 commute like translations and rotate into each other under J_3 . The little group of a massless particle in $3 + 1$ dimensions is therefore

$$G_{\text{massless}} = ISO(2), \quad (68)$$

the Euclidean group $ISO(n - 2)$ with $n = 4$ (E. Wigner, “On Unitary Representations of the Inhomogeneous Lorentz Group”, *Annals of Mathematics* **40** (1939) 149–204, DOI: 10.2307/1968551; S. Weinberg, *The Quantum Theory of Fields*, Vol. 1, Cambridge University Press, 1995, §2.5).

10.2 Spin for massive particles, helicity for massless ones

Massive. The unitary representations of $SO(3)$ — more precisely of its double cover $SU(2)$, which is what acts on quantum states — are the familiar spin multiplets: for each $s \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ there is exactly one irreducible representation, of dimension

$$\dim = 2s + 1, \quad (69)$$

labeled by the eigenvalues $-s, -s + 1, \dots, +s$ of any one rotation generator (e.g. J_3). The restriction to integer and half-integer s — as opposed to the real-valued s allowed in $2 + 1$ dimensions — comes from the topology of the group: $\pi_1(SO(3, 1)) = \mathbb{Z}_2$, so only the double cover is needed, and $SU(2)$ has only these representations (Weinberg, loc. cit., §2.7; the π_1 statement follows from the standard deformation retract of $SO(3, 1)$ onto its maximal compact subgroup $SO(3)$, with $\pi_1(SO(3)) = \mathbb{Z}_2$ tabulated in M. Nakahara, *Geometry, Topology and Physics*, IOP, 2nd ed. 2003, §4.7, Table 4.1).

Massless. The little group $ISO(2)$ has two kinds of unitary irreducible representations. In the first kind the “translation” generators T_1, T_2 act non-trivially, producing a continuous infinity of internal states — the **continuous-spin representations**. No such particle has ever been observed, and the standard physical assumption, going back to Wigner’s 1939 paper, is that T_1 and T_2 act trivially on physical states. What remains is a representation of the residual $SO(2)$ generated by J_3 : a single number

$$J_3 \mapsto \lambda, \quad \lambda \in \{0, \pm\frac{1}{2}, \pm 1, \dots\}, \quad (70)$$

the **helicity** — the spin component along the direction of motion. The allowed values are again quantized to integers and half-integers by the $\mathbb{Z}_2$ topology. (The helicity is Lorentz-invariant for a massless particle: no Lorentz transformation can reverse the direction of motion relative to the spin without leaving the null cone, which is why a single λ labels the whole orbit. This is the group-theoretic content of (66)–(68); see Weinberg, §2.5.)

A massless *field* furthermore comes with both helicities $\pm|\lambda|$: the discrete spacetime symmetries relate the two, and any local massless field theory that respects them contains both polarizations (Weinberg, §5.9, where this is the starting point of the construction of the electromagnetic and gravitational fields). The photon is the case $|\lambda| = 1$: two polarization states. The graviton is the case $|\lambda| = 2$: two polarization states.

10.3 The smooth classical limit and the non-smooth state counting

We can now state the reconciliation that the whole construction was aiming at.

- **Classically smooth.** The einbein action (10) depends analytically on m through the single term $-\frac{\epsilon}{2}(mc_0)^2$. At $m = 0$ nothing degenerates: the variational principle, the constraint, and the geodesic equation all persist, and massive trajectories approach null geodesics as the initial velocity is pushed onto the cone. This is the content of Sections 5–9, and it is dimension-independent.

- **Kinematically continuous, with a change of type.** The celestial sphere S^2 is the ideal boundary of the mass shell H^3 (hyperboloid model; Ratcliffe, loc. cit.). Every null direction is a limit of timelike directions and conversely. What changes type at $m = 0$ is the stabilizer: $SO(3)$ for every $m > 0$, $ISO(2)$ at $m = 0$ exactly.
- **Quantum-mechanically non-smooth.** The internal-state counting jumps: $2s+1$ massive spin states versus two massless helicity states (one helicity value λ for a particle, paired to $\pm|\lambda|$ in any massless field). This is Wigner kinematics, not a dynamical pathology — but for *interacting* theories it has dynamical teeth, as the next subsection records.

10.4 The vDVZ discontinuity: when the state-counting jump becomes observable

For free particles the loss of states at $m = 0$ is invisible in the classical limit. For interacting massive spin-2 (a massive graviton), the extra states do not decouple as $m \rightarrow 0$: the helicity-0 mode continues to couple to the trace of the energy-momentum tensor with finite strength, so the $m \rightarrow 0$ limit of the massive theory’s *predictions* (e.g. the bending of light by a massive source) differs from the massless theory’s by a finite factor — the celebrated result that massive gravity in the limit of vanishing graviton mass does not reduce to general relativity (H. van Dam and M. J. G. Veltman, “Massive and mass-less Yang–Mills and gravitational fields”, *Nuclear Physics B* **22** (1970) 397–411, DOI: 10.1016/0550-3213(70)90416-5; V. I. Zakharov, “Linearized gravitation theory and the graviton mass”, *JETP Letters* **12** (1970) 312–314; a modern review is K. Hinterbichler, “Theoretical aspects of massive gravity”, *Reviews of Modern Physics* **84** (2012) 671–710, DOI: 10.1103/RevModPhys.84.671). The discontinuity is precisely the interacting shadow of the little-group jump $SO(3) \rightarrow ISO(2)$ of Section 10.1. For spin 1, by contrast, the extra longitudinal mode decouples from conserved currents as $m \rightarrow 0$, and the limit is smooth (van Dam–Veltman, loc. cit.) — current conservation, not kinematics alone, decides whether the state-counting jump is observable.

This tutorial’s subject is the free particle, where the einbein delivers a perfectly smooth massless limit; Sections 10.2–10.4 exist to mark exactly how far that smoothness extends, and where it ends.

11 Brief Hamiltonian Perspective

For completeness we record the Hamiltonian formulation that follows from the einbein Lagrangian. The canonical momenta conjugate to the coordinates are

$$p_\lambda = \frac{\partial L}{\partial \dot{x}^\lambda} = \frac{1}{e} g_{\lambda\nu} \dot{x}^\nu. \quad (71)$$

Inverting for the velocities, $\dot{x}^\nu = e g^{\nu\lambda} p_\lambda$. The Hamiltonian (after a Legendre transform) is pure constraint:

$$H = \frac{e}{2} (g^{\mu\nu} p_\mu p_\nu + (mc_0)^2). \quad (72)$$

The einbein itself acts as a Lagrange multiplier enforcing the mass-shell condition

$$g^{\mu\nu} p_\mu p_\nu + (mc_0)^2 = 0. \quad (73)$$

In an inertial frame this reads $p_0^2 = |\mathbf{p}|^2 + (mc_0)^2$, and in the massless case $p_0 = |\mathbf{p}|$ — with $p_0 = E/c_0$ the familiar $E = |\mathbf{p}| c_0$ of the photon. In $3 + 1$ dimensions the massive shell is the hyperbolic three-space H^3 of Section 10.1 and the massless shell is the punctured null cone; both are three-dimensional surfaces in the four-dimensional momentum space, so the constraint surface keeps its dimension across the limit while changing its geometry and its stabilizer structure. This constrained Hamiltonian system is the starting point for the canonical quantization of the relativistic particle, and the little-group representation theory of Section 10 is the quantum theory of the degrees of freedom that remain after the constraint (73) is imposed.

12 Resolution of the Equations of Motion and the Resulting Trajectories

After the residual reparametrization freedom has been used to set $e = \text{const}$, the dynamical content of the theory is completely determined.

12.1 Massive particle ($m > 0$)

In the gauge $e = \text{const}$ the Euler–Lagrange equations reduce to the pair

$$\frac{D\dot{x}^\sigma}{d\tau} = \ddot{x}^\sigma + \Gamma^\sigma_{\rho\nu} \dot{x}^\rho \dot{x}^\nu = 0, \quad (74)$$

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -e^2 (mc_0)^2 = \text{const} < 0. \quad (75)$$

Well-posedness. Given data $x^\mu(0) = x_0^\mu$, $\dot{x}^\mu(0) = u_0^\mu$ satisfying the mass-shell condition at $\tau = 0$, the Picard–Lindelöf theorem guarantees a unique local solution; the constraint is preserved by (74) because the Levi-Civita connection is metric-compatible: $\frac{d}{d\tau} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu) = 2 g_{\mu\nu} \dot{x}^\mu \frac{D\dot{x}^\nu}{d\tau} = 0$ (verified numerically in the companion suite). The solution extends to a maximal interval and is global if the spacetime is timelike-geodesically complete.

Free data. The constraint (75) fixes only the overall scale of the tangent vector (its norm is pinned to $-e^2(mc_0)^2$); the three independent ratios of the four velocity components — two direction angles (ϑ_0, φ_0) and the speed ratio — are free initial data. Each Killing vector field⁷ of the metric supplies one conserved quantity $E = g_{\mu\nu} \xi^\mu \dot{x}^\nu$ and reduces the system accordingly.

Flat-space illustration. When $g_{\mu\nu} = \eta_{\mu\nu}$ the Christoffel symbols vanish and the equations become $\ddot{x}^\sigma = 0$ with $\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -e^2(mc_0)^2$. The general solution is the straight line

$$x^\sigma(\tau) = x_0^\sigma + u_0^\sigma \tau, \quad \eta_{\mu\nu} u_0^\mu u_0^\nu = -e^2(mc_0)^2, \quad (76)$$

the exponential map of a vector space.

12.2 Massless particle ($m = 0$)

In the same gauge the equations reduce to

$$\frac{D\dot{x}^\sigma}{d\tau} = 0, \quad g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0. \quad (77)$$

Any solution is a null geodesic of the background metric, affinely parametrized. The initial direction of the null vector — a point of the celestial sphere S^2 of Section 9.3 — is part of the data. In Minkowski space the solutions are the straight null lines

$$x^\sigma(\tau) = x_0^\sigma + k_0^\sigma \tau, \quad \eta_{\mu\nu} k_0^\mu k_0^\nu = 0, \quad k_0^0 > 0. \quad (78)$$

As in every dimension, the *unparametrized* null geodesics depend only on the conformal class of the metric: under a Weyl rescaling $g_{\mu\nu} \rightarrow \Omega^2(x) g_{\mu\nu}$ the null cone is unchanged pointwise and the null geodesic equation is preserved up to reparametrization (R. M. Wald, *General Relativity*, Appendix D). At the level of the action the einbein makes this transparent: under the combined transformation

$$g_{\mu\nu} \mapsto \Omega^2(x) g_{\mu\nu}, \quad e \mapsto \Omega^2(x(\tau)) e(\tau) \quad (79)$$

⁷A Killing vector field is a vector field ξ^μ along whose flow the metric is invariant, equivalently $\nabla_{(\mu} \xi_{\nu)} = 0$. The inner product of a Killing vector with the tangent vector of a geodesic is constant along the geodesic; see R. M. Wald, *General Relativity* (University of Chicago Press, 1984), Appendix C, or Carroll, *Spacetime and Geometry*, Appendix B.

the massless Lagrangian (56) is invariant pointwise, while the mass term $-\frac{\epsilon}{2}(mc_0)^2$ picks up a factor Ω^2 (both statements checked symbolically in the companion suite). This conformal invariance is the classical root of the conformal symmetry of the free Maxwell and Yang–Mills actions in $3 + 1$ dimensions — the field-theoretic descendants of the massless particle — though establishing the field-theoretic statement is beyond our present scope.

12.3 Summary of the trajectories

Case	Final equations	Geometric character	Free direction data
Massive ($m > 0$)	affine geodesic eq. + mass shell	future-directed timelike geodesic	$(\vartheta_0, \varphi_0) \in S^2$ plus speed ratio
Massless ($m = 0$)	affine geodesic eq. + null condition	null geodesic	$(\vartheta_0, \varphi_0) \in S^2$

13 Summary

We have shown, by completely explicit algebraic manipulations, that the einbein reformulation of the relativistic point particle extends from $1 + 1$ to $3 + 1$ dimensions without any modification of its variational core: the auxiliary-field equation (25), the recovery of the square-root action (Section 7), the geodesic equation with its non-affine term (50), and the gauge $e = \text{const}$ that removes it are all dimension-independent. The new content of the $3 + 1$ -dimensional theory is kinematic and group-theoretic:

- the null directions form the celestial sphere S^2 (Section 9.3);
- the massive mass shell is the hyperbolic three-space H^3 with little group $SO(3)$ and spin multiplets of dimension $2s + 1$ (Sections 10.1–10.2);
- the massless little group is the Euclidean group $ISO(2)$, exhibited explicitly by the generators (65) and their algebra (67); physical massless states carry a single helicity λ , paired as $\pm|\lambda|$ in any massless field — two polarization states for the photon and the graviton (Section 10.2);
- the massless limit is smooth at the level of the action and the geodesics, while the state counting jumps from $2s + 1$ to two; for interacting massive spin-2 the jump is observable as the van Dam–Veltman–Zakharov discontinuity (Sections 10.3–10.4);
- the massless einbein action is Weyl-invariant, exposing the conformal nature of unparametrized null geodesics (Section 12.2).

With the $1 + 1$ construction of [0], the $2 + 1$ sequel, and the present notes, the einbein formulation of the relativistic particle is now documented in the full dimensional ladder, with every derivable claim backed by an explicit machine check.

Appendix A. The Forty Independent Christoffel Components of a General $3 + 1$ Metric

The forty independent components of the Christoffel symbol (49) in a general $3 + 1$ -dimensional metric, organized by the upper index σ and the symmetric lower pair (ρ, ν) with $\rho \leq \nu$. This listing is **generated programmatically** from equation (49) by `generate_christoffel_block.py` (same directory as this document); the companion test suite regenerates the block, diffs it against the text below, and separately checks every component numerically against the general formula on random metric data. The components with $\rho > \nu$ follow from $\Gamma^\sigma_{\rho\nu} = \Gamma^\sigma_{\nu\rho}$.

For $\sigma = 0$:

$$\begin{aligned}
\Gamma^3{}_{00} &= \frac{1}{2}g^{30}\partial_0g_{00} + \frac{1}{2}g^{31}(2\partial_0g_{10} - \partial_1g_{00}) + \frac{1}{2}g^{32}(2\partial_0g_{20} - \partial_2g_{00}) + \frac{1}{2}g^{33}(2\partial_0g_{30} - \partial_3g_{00}) \\
\Gamma^3{}_{01} &= \frac{1}{2}g^{30}\partial_1g_{00} + \frac{1}{2}g^{31}\partial_0g_{11} + \frac{1}{2}g^{32}(\partial_0g_{21} + \partial_1g_{20} - \partial_2g_{01}) + \frac{1}{2}g^{33}(\partial_0g_{31} + \partial_1g_{30} - \partial_3g_{01}) \\
\Gamma^3{}_{02} &= \frac{1}{2}g^{30}\partial_2g_{00} + \frac{1}{2}g^{31}(\partial_0g_{12} + \partial_2g_{10} - \partial_1g_{02}) + \frac{1}{2}g^{32}\partial_0g_{22} + \frac{1}{2}g^{33}(\partial_0g_{32} + \partial_2g_{30} - \partial_3g_{02}) \\
\Gamma^3{}_{03} &= \frac{1}{2}g^{30}\partial_3g_{00} + \frac{1}{2}g^{31}(\partial_0g_{13} + \partial_3g_{10} - \partial_1g_{03}) + \frac{1}{2}g^{32}(\partial_0g_{23} + \partial_3g_{20} - \partial_2g_{03}) + \frac{1}{2}g^{33}\partial_0g_{33} \\
\Gamma^3{}_{11} &= \frac{1}{2}g^{30}(2\partial_1g_{01} - \partial_0g_{11}) + \frac{1}{2}g^{31}\partial_1g_{11} + \frac{1}{2}g^{32}(2\partial_1g_{21} - \partial_2g_{11}) + \frac{1}{2}g^{33}(2\partial_1g_{31} - \partial_3g_{11}) \\
\Gamma^3{}_{12} &= \frac{1}{2}g^{30}(\partial_1g_{02} + \partial_2g_{01} - \partial_0g_{12}) + \frac{1}{2}g^{31}\partial_2g_{11} + \frac{1}{2}g^{32}\partial_1g_{22} + \frac{1}{2}g^{33}(\partial_1g_{32} + \partial_2g_{31} - \partial_3g_{12}) \\
\Gamma^3{}_{13} &= \frac{1}{2}g^{30}(\partial_1g_{03} + \partial_3g_{01} - \partial_0g_{13}) + \frac{1}{2}g^{31}\partial_3g_{11} + \frac{1}{2}g^{32}(\partial_1g_{23} + \partial_3g_{21} - \partial_2g_{13}) + \frac{1}{2}g^{33}\partial_1g_{33} \\
\Gamma^3{}_{22} &= \frac{1}{2}g^{30}(2\partial_2g_{02} - \partial_0g_{22}) + \frac{1}{2}g^{31}(2\partial_2g_{12} - \partial_1g_{22}) + \frac{1}{2}g^{32}\partial_2g_{22} + \frac{1}{2}g^{33}(2\partial_2g_{32} - \partial_3g_{22}) \\
\Gamma^3{}_{23} &= \frac{1}{2}g^{30}(\partial_2g_{03} + \partial_3g_{02} - \partial_0g_{23}) + \frac{1}{2}g^{31}(\partial_2g_{13} + \partial_3g_{12} - \partial_1g_{23}) + \frac{1}{2}g^{32}\partial_3g_{22} + \frac{1}{2}g^{33}\partial_2g_{33} \\
\Gamma^3{}_{33} &= \frac{1}{2}g^{30}(2\partial_3g_{03} - \partial_0g_{33}) + \frac{1}{2}g^{31}(2\partial_3g_{13} - \partial_1g_{33}) + \frac{1}{2}g^{32}(2\partial_3g_{23} - \partial_2g_{33}) + \frac{1}{2}g^{33}\partial_3g_{33}
\end{aligned}$$

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End of tutorial notes.